Manu Jayadharan, PhD

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Research Interests

Stable and interpretable algorithms for forward and inverse scientific computing problems. On the forward side: finite-element methods, domain decomposition, and multiscale discretizations for multiphysics PDEs (Biot poroelasticity, Stokes–Biot, Poisson–Nernst–Planck). On the inverse side: data-driven discovery of differential and differential-algebraic equations, sparse optimization, well-posedness and identifiability of dictionary-based model-discovery methods, multiple-shooting parameter inference, and hybrid mechanistic–ML modelling. Open-source scientific software as a first-class research output.

Education

University of Pittsburgh, Pittsburgh, PA, USA — Aug 2016 – Aug 2021
Ph.D. in Mathematics
Advisor: Prof. Ivan Yotov
Dissertation: *Domain Decomposition and Time-Splitting Methods for the Biot System of Poroelasticity*

IISER Mohali, Mohali, India — Aug 2011 – May 2016
BS-MS Dual Degree in Mathematics
DST INSPIRE Fellow; Top rank holder, Department of Mathematics
MS Thesis: *Study of Cauchy's Basic Equations and Convex Functions*

Academic Positions

Northwestern University, Evanston, IL, USA — Sept 2023 – Present
Postdoctoral Fellow, Engineering Sciences & Applied Mathematics (PI: Prof. Niall Mangan)
Affiliated researcher, National Institute for Theory and Mathematics in Biology (NITMB, NSF–Simons-funded centre at Northwestern and UChicago); Trienens Institute for Sustainability and Energy at Northwestern.

- Developed numerically stable, data-driven algorithms for discovering differential and differential-algebraic equations from noisy observational data, motivated by applications in power-grid modelling, chemical engineering, and battery design.
- Identified and analysed structural ill-posedness in dictionary-based model-discovery methods, linked it to inverse-problem theory, and proposed QR-based stabilization.
- Led design and open-source release of `dae-finder`, a model-agnostic Python package for discovering differential-algebraic equations via sparse optimization (PyPI).
- Lead developer of fast, flexible Galerkin weak-form solvers for the coupled Poisson–Nernst–Planck equations with stiff Butler–Volmer boundary conditions (electrochemical reactor modelling, Trienens Institute).
- Contributed to grant proposals on fast solvers for multiphysics PDEs and on data-driven PDE

discovery; regular research reporting to NITMB and Trienens Institute stakeholders.

Industry Experience

Citigroup Inc., New York, NY, USA May 2021 – Aug 2023

Quantitative Analyst, AVP, Global Credit Quantitative Analysis (LATAM Emerging Markets Desk)

FINRA-licensed; production C++ analytics library; risk and sensitivity measures; model validation.

- Contributed to large-scale quantitative libraries for pricing financial derivatives and computing risk and sensitivity measures used by Citi's trading desks worldwide.
- Implemented new risk metrics; supported model validation and deployment in a complex legacy production C++ codebase with tens of thousands of regression tests.
- Built two production tools used on the LATAM EM credit desk: *Flash P&L* analysis tool (portfolio risk breakdown with parallel computation) and *ExtPricer* (extended interest-rate-swap pricing engine).
- The model-agnostic architecture of Citi's pricing library directly informed the design of `dae-finder`.

Publications

Refereed Journal Articles: Published / Accepted

1. **M. Jayadharan**, N. M. Mangan et al., "SODAs: Sparse Optimization for Discovery of Differential-Algebraic Systems from Data," *Proceedings of the Royal Society A*, 2026. DOI: 10.1098/rspa.2025.0201.

2. **M. Jayadharan**, I. Yotov, "Multiscale mortar mixed finite element methods for the Biot system of poroelasticity," *Computer Methods in Applied Mechanics and Engineering*, Vol. 435, Art. 117597, 2025.

3. **M. Jayadharan**, M. Kern, M. Vohralík, I. Yotov, "A space-time multiscale mortar mixed finite element method for parabolic equations," *SIAM Journal on Numerical Analysis*, 2023.

4. **M. Jayadharan**, E. Khattatov, I. Yotov, "Domain decomposition and partitioning methods for mixed finite element discretizations of the Biot system of poroelasticity," *Computational Geosciences*, 25, 1919–1938, 2021.

Under Review / Preprint

5. Y. Feng, N. M. Mangan, **M. Jayadharan**[†], "Ill-conditioning in dictionary-based dynamic-equation learning: A systems biology case study," arXiv:2603.11330, 2026. (Under review, *SIAM Journal on Life Sciences*.)

In Preparation

6. H. Chen, N. M. Mangan, **M. Jayadharan**[†], "Improving Extrapolation in Neural ODEs via Monomial Augmentation and Symbolic Structure Recovery," 2026.

7. **M. Jayadharan**, N. M. Mangan, "Stabilizing Sparse Model Identification through Truly Orthogonal Candidate Libraries," 2026.

8. **M. Jayadharan**, N. M. Mangan, "A Multi-Shooting Framework for Sparse Model Discovery with Guess Propagation," 2026.

9. M. Gumman, **M. Jayadharan**, I. Yotov, "Domain decomposition methods for the Stokes–Biot model of fluid–poroelastic structure interaction," 2026.

† *corresponding / senior author. Senior-author papers (5, 7) reflect an independent research direction developed at Northwestern on stability and structural recovery in data-driven equation learning.*

Selected Ongoing Projects

- **Agentic AI for Scientific Computing** (active 2025–present). Building protocols, validation frameworks, and reusable agentic skill sets for the use of frontier AI agents (Claude Code, Codex / GPT, Gemini) in scientific computing, including numerical solvers for stiff nonlinear ODEs / PDEs, parameter estimation, and HPC orchestration. Anchor application: a coupled Poisson–Nernst–Planck inverse problem with downstream applications in chemical engineering and battery materials. Collaborative with members of the Center for Optimization and Statistical Learning at Northwestern (incl. discussions with Jorge Nocedal's group). Initial grant funding secured. First graduate course offering, *Agentic AI for Scientific Computing*, at Northwestern Applied Mathematics, Fall 2026.

Open-Source Scientific Software

- **DAE-FINDER** (`dae-finder` on PyPI). Model-agnostic Python package for discovering differential-algebraic equations from noisy data via sparse optimization; supports any algorithm exposing a `.fit()`/`.score()` interface. Released and maintained.
- **FluidLearn**. Physics-informed neural networks for multiphysics PDEs (Python / TensorFlow/Keras).
- **SpaceTimeDD**. Parabolic-PDE solver via space-time domain decomposition (C++).
- **BiotDDSolver**. MPI-based domain-decomposition simulator for Biot poroelasticity (C++, MPI).
- GitHub: mjayadharan.github.io.

Invited Talks

- *From Data to Differential Equations*, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, August 2025. Host: Prof. Deepak Mishra.
- *SODAs for Discovering Differential-Algebraic Equations*, SIAM Annual Meeting AN25, July 2025.
- *On discovery of differential-algebraic equations*, SIAM Conference on Mathematics in Data Science, October 2024.

Conference Presentations and Service

- *On numerical methods for discovering differential equations*, SIAM Annual Meeting AN24, July 2024.
- Organizer, Minisymposium on *Numerical Methods for Sparse Optimization*, SIAM AN24.
- *Domain decomposition and partitioning methods for mixed FEM of the Biot system of poroelasticity*, SIAM Conference on Mathematical and Computational Issues in the Geosciences, Milan, June 2021.
- *Discretization techniques for the Biot system of poroelasticity*, FE Circus, Blacksburg, VA, November 2019.

Teaching Experience

- **Northwestern University** Fall 2026
  Instructor, *Agentic AI for Scientific Computing* (new graduate course, Applied Mathematics; project-based, with funded frontier-model access for enrolled students).
- **Northwestern University** Fall 2023
  Instructor, Engineering Analysis 4: Differential Equations and Numerical Methods.
- **University of Pittsburgh** Summer 2017
  Instructor, MATH 290: Introduction to Ordinary Differential Equations and Applications.
- **University of Pittsburgh** 2016 – 2021
  Teaching Fellow / Teaching Assistant: Calculus I–III, Business Calculus, Ordinary Differential Equations.

Mentoring

Mentored six undergraduate and two graduate / master's students across Northwestern University, UC Berkeley, Carnegie Mellon University, IIST (Thiruvananthapuram), and the University of Pittsburgh. Several mentees are co-authors on papers in preparation: in particular, my Northwestern mentee Henry Chen is first author and I am senior / corresponding author on the in-preparation monomial-augmented neural ODE manuscript; graduate student Gumman at Pittsburgh is first author on the in-preparation Stokes–Biot domain decomposition paper I co-supervise with Prof. Ivan Yotov.

Awards, Fellowships, and Honors

- Provost's Dissertation Completion Fellowship, University of Pittsburgh, 2021.
- Graduate Student Researcher (NSF support), University of Pittsburgh, 2018–2020.
- Arts & Sciences Graduate Fellowship, University of Pittsburgh, 2017.
- UGC–CSIR NET and IIT–GATE in Mathematics, 2016.
- Summer Research Intern Fellowship, NNMCB, IISc Bengaluru, 2015.
- Merit Certificate for Academic Excellence in Mathematics, IISER Mohali, 2014.
- DST INSPIRE Scholarship, Government of India, 2011–2016.
- Top 0.1% merit certificate in Physics, national CBSE examination, 2011.

Professional Service

- Affiliated researcher, NITMB (NSF–Simons Institute for Mathematics in Biology at Northwestern / UChicago), current.
- Vice President, Association for Women in Mathematics (Pitt Chapter), 2020–2021.
- Officer, Graduate Student Organization, University of Pittsburgh, 2018–2020.
- Organizer, Minisymposium on Numerical Methods for Sparse Optimization, SIAM AN24, 2024.

Technical Skills

Languages: Python, C++, Julia, MATLAB, Fortran.

Numerical / SciML: FEM (deal.II, FEniCS, FreeFem++), sparse optimization, neural ODEs, SINDy / DAE-FINDER family, PINNs, multiple shooting, QR orthogonalization, domain decomposition, weak-form and variational methods.

HPC: MPI, Slurm, cluster computing.

ML / Data: NumPy, SciPy, SymPy, scikit-learn, TensorFlow / Keras, Pyro.

Application domains: Applicable mathematics, numerical PDEs, electrochemistry, systems biology, power-grid modelling, quantitative finance.

References

Available upon request.